Let (X, τ) be the topological space and A,B⊂X such that A⊂B then for derived set A' and

Select the correct option

PRESENTED BY: TALEEMI MARKAZ

- $(A \cup B)' = A' \cup B'$ (Lecture 7, Topic 23)
- $(A \cup B)' = A' \cap B'$
- $(A \cup B)' = A \cup B$
- None of above

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Question # 1 of 30

SOLVED BY: WAQAR HUSSAIN

For any set there are always two topologies namely _____

Select the correct option

- Lower li
- Discrete
- Discrete
- Co finite and indiscrete Topology

Some Special Topologies

For any set there are always two topologies those are very special.

1. Indiscrete Topology.
2. Discrete Topology:

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Question # 2 of 30

SOLVED BY: WAQAR HUSSAIN

Topology means the study of something with respect to its _____

Select the correct option

- ○ volume
- ○ perimeter
- ○ place
- ○ area

Topology as a word is composition of two words

1. Topos(Greek) means "Place".
2. Logy(Latin) means "Study/Analysis".

The study of some thing with respect to its place.

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Question # 3 of 30

SOLVED BY: WAQAR HUSSAIN

Which of the following is true for a subset A of a Topological space X.

Select the correct option

- $X = A^0 \cup Ext(A) \cup bd(A)$
- $X = A^0 \cap Ext(A) \cup bd(A)$
- $X = A^0 \cap Ext(A) \cap bd(A)$
- $X = A^0 \cup Ext(A) \cap bd(A)$

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Question # 4 of 30

SOLVED BY: WAQAR HUSSAIN

The lower limit topology is _____ than the usual topology.

Select the correct option

- finer
- coarser
- equal
- none of these

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Question # 6 of 30

SOLVED BY: WAQAR HUSSAIN

Only one topology can be defined on a set

Select the correct option

- ○ True
- ○ False
- ○
- ○

Remark 1:
More than one topologies can be defined on a set.

Let us define all the possible topologies on 1-point, 2-point and 3-point sets.

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Question # 7 of 30

SOLVED BY: WAQAR HUSSAIN

The smallest topology one can define on some set is called

Select the correct option

- co
- indi
- discrete topology
- usual topology

Some Special Topologies

Remark:
Indiscrete topology is the smallest topology one can define on some set.

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Question # 8 of 30

SOLVED BY: WAQAR HUSSAIN

Which of the following is the closed super set of a subset A of a topological space X?

Select the correct option

- Derived set
- Complemer
- Closure of
- open sets of A

Closure of a Set

Remarks:

- $A \subset \overline{A}$
- $\overline{A}$ is the smallest closed superset of A.

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Question # 8 of 30

SOLVED BY: WAQAR HUSSAIN

Which of the following is the closed super set of a subset A of a topological space X?

Select the correct option

- Derived set of A
- Complement of A
- Closure of a set A
- open sets of A

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Question # 9 of 30

SOLVED BY: WAQAR HUSSAIN

In a discrete space X, the derived set A' of any subset A of X is _____

Select the correct option

- A it
- A cc
- emp ,
- X

Derived Set: Special Topologies

Following is our concluding remark.

Remark:

In a discrete space X, the derived set A' of any subset A of X is empty.

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Question # 9 of 30

SOLVED BY: WAQAR HUSSAIN

In a discrete space X, the derived set A' of any subset A of X is _____

Select the correct option

- ○ A itself
- ○ A complement
- ● empty set
- ○ X

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Question # 10 of 30

SOLVED BY: WAQAR HUSSAIN

Which of the following statement is true?

Select the correct option

- In Geon ... we don't
- In both ... rements
- In Geon ... ology we
- In both ... measurements

Introduction to Topology

In Geometry we care about the exact measurements such as lengths, angles and areas.

But in Topology we don't care about the measurements.

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Question # 11 of 30

SOLVED BY: WAQAR HUSSAIN

If one shape can be deformed in another shape then topologically they are considered to be _____

Select the correct option

- the same
- dissimilar
- unique
- partially equal

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Question # 13 of 30

SOLVED BY: WAQAR HUSSAIN

Which of the following statement is not true?

Select the correct option

- Indiscrete topology is weaker than any other topology defined on the same non empty set
- Discrete topology is finer than any other topology defined on the same non empty set
- Discrete topology is finer than the indiscrete topology defined on the non empty set
- Indiscrete topology is finer than any topology defined on the same non empty set

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Question # 14 of 30

SOLVED BY: WAQAR HUSSAIN

Which of the following is the basis for an upper limit topology on R (The Set of Real Numbers)?

Select the correct option

- The Upper Limit Topology or R

  Of course, we can also define the upper limit topology on $\mathbb{R}$ analogously as follows:

  Definition: The **Upper Limit Topology** on the set of real numbers $\mathbb{R}$, τ is the topology generated by all unions of intervals of the form $\{(a, b] : a, b \in \mathbb{R}, a \leq b\}$.

- $B_{up} = \{[a, b] \mid a, b \in R\}$

- $B_{up} = \{[a, b) \mid a, b \in R\}$

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Question # 14 of 30

SOLVED BY: WAQAR HUSSAIN

Which of the following is the basis for an upper limit topology on R (The Set of Real Numbers)?

Select the correct option

- $B_{up} = \{(a,b) \mid a,b \in R\}$
- $B_{up} = \{(a,b] \mid a,b \in R\}$
- $B_{up} = \{[a,b] \mid a,b \in R\}$
- $B_{up} = \{[a,b) \mid a,b \in R\}$

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Question # 15 of 30

SOLVED BY: WAQAR HUSSAIN

The largest topology defined on some set is the _____ topology.

Select the correct option

- indiscrete
- lower limit
- co finite
- discrete

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Question # 18 of 30

SOLVED BY: WAQAR HUSSAIN

Interior of a set A (denoted by Ao) which is a subset of a topological space X is the ____ of all ____ subsets of X contained in A

Select the correct option

- intersection, open
- union, closed
- intersection, closed
- union, open

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Question # 17 of 30

SOLVED BY: WAQAR HUSSAIN

Exterior point of a set is the _____ point of its complement.

Select the correct option

- interior
- boundary
- exterior
- limit

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Question # 16 of 30

SOLVED BY: WAQAR HUSSAIN

Which of the following are NOT topologically equal?

Select the correct option

- A circle and a square
- A triangle and a rectangle
- A rectangle and a square
- A circle and a line

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Question # 19 of 30

SOLVED BY: WAQAR HUSSAIN

_____ is the largest topology containing both T1 and T2.

Select the correct option

- T1 intersection T2
- T1 union T2
- Cofinite topology
- Indiscrete topology

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Question # 20 of 30

SOLVED BY: WAQAR HUSSAIN

The set of _____ of R (Real line) forms a topology called usual topology.

Select the correct option

- ○ all open discs
- ○ all open balls
- ◔ all open intervals
- ○ all open spheres

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Question # 21 of 30

SOLVED BY: WAQAR HUSSAIN

Which of the following is the basis for a lower limit topology on R (The Set of Real Numbers)?

Select the correct option

- $B_l = \{[a,b) \mid a,b \in R\}$
- $B_l = \{(a,b) \mid a,b \in R\}$
- $B_l = \{[a,b] \mid a,b \in R\}$
- $B_l = \{(a,b] \mid a,b \in R\}$

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Question # 24 of 30

SOLVED BY: WAQAR HUSSAIN

Which of the following represents an open interval on real line R?

Select the correct option

- $(x|a < x < b)$ where $a, b \in R$ with usual order $a < b$
- $(x|a \leq x < b)$ where $a, b \in R$ with usual order $a < b$
- $(x|a \leq x \leq b)$ where $a, b \in R$ with usual order $a < b$
- $(x|a < x \leq b)$ where $a, b \in R$ with usual order $a < b$

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Question # 27 of 30

SOLVED BY: WAQAR HUSSAIN

Boundary point of a set is also called _____ point.

Select the correct option

- exterior
- interior
- frontier
- limit

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Question # 23 of 30

SOLVED BY: WAQAR HUSSAIN

Closure of a subset A of a topological space X is the intersection of all _____ of X containing A

Select the correct option

- open subsets
- subsets
- closed subsets
- none of these

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Question # 26 of 30

SOLVED BY: WAQAR HUSSAIN

The set of all singletons forms a basis for _____ topology.

Select the correct option

- discrete
- indiscrete
- co finite
- lower limit

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Question # 25 of 30

SOLVED BY: WAQAR HUSSAIN

If X is a finite set then co finite topology on X is _____

Select the correct option

- indiscrete topology
- lower limit topology
- none of these
- discrete topology

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Question # 22 of 30

SOLVED BY: WAQAR HUSSAIN

A function is continuous _____ it is continuous at ___ point of the domain space.

Select the correct option

- if, every
- if and only if, every
- if and only if, any
- if, any

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Question # 28 of 30

SOLVED BY: WAQAR HUSSAIN

The basic for usual topology on R is the set of all ____

Select the correct option

- open intervals
- closed intervals
- open disks
- closed disks

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Question # 29 of 30

SOLVED BY: WAQAR HUSSAIN

If B is a basis for a topology T on a set X then _____ element of T can be written as union of the elements of B

Select the correct option

- every
- no
- some
- one

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Question # 30 of 30

SOLVED BY: WAQAR HUSSAIN

Continuity of a function in a topological space depends on the collection of _____ of both sets.

Select the correct option

- open subsets
- closed subsets
- proper subsets
- Improper subsets

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 4 of 30 (Start time: 09:07:53 AM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d\}$. *The following set represents a topology on* X.

Select the correct option

Reload Math Equations

- {φ,{c},{a,b},X}
- {φ,{a},{b,c},X}
- {φ,{a},{b},X}
- {φ,{a},{a,b},X}

Click to Save Answer & Move to Next Question

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 3 of 30 (Start time: 09:06:16 AM, 02 July 2021)

Total Marks: 1

Let $X = \mathbb{R}$ with usual topology and $A = (0, 3)$. *The limit point of* A is:

Select the correct option

Reload Math Equations

- 5
- 4
- -0 5
- 1 5

Click to Save Answer & Move to Next Question

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 1 of 30 (Start time: 09:02:06 AM, 02 July 2021)

Total Marks: 1

If τ is a topology on a non empty set X, then arbitrary ..... of members of τ belongs to τ.

Select the correct option

Reload Math Equations

- union
- intersection

Click to Save Answer & Move to Next Question

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 5 of 30 (Start time: 09:08:26 AM, 02 July 2021)

Total Marks: 1

In a topological space, the interior of a set A is the

Select the correct option

Reload Math Equations

- smallest open subset of A
- largest open subset of A

Click to Save Answer & Move to Next Question

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 6 of 30 (Start time: 09:10:30 AM, 02 July 2021)

Total Marks: 1

Which of the following is the basis for a lower limit topology on R (The Set of Real Numbers)?

Select the correct option

Reload Math Equations

- $B_l = \{[a, b) \mid a, b \in R\}$
- $B_l = \{[a, b] \mid a, b \in R\}$
- $B_l = \{(a, b) \mid a, b \in R\}$
- $B_l = \{(a, b] \mid a, b \in R\}$

Click to Save Answer & Move to Next Question

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 8 of 30 (Start time: 09:14:04 AM, 02 July 2021)

Total Marks: 1

The set of all open intervals of $\mathbb{R}$ is a topology on $\mathbb{R}$, called

Select the correct option

Reload Math Equations

- discrete topology
- usual topology
- cofinite topology
- real topology

Click to Save Answer & Move to Next Question

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 12 of 30 (Start time: 09:18:57 AM, 02 July 2021)

Total Marks: 1

Let (X, τ) be the topological space and $A, B \subset X$ such that $A \subset B$ then for derived set

$$A'$$

and

$$B'$$

Select the correct option

Reload Math Equations

- $(A \cup B)' = A' \cup B'$
- $(A \cup B)' = A \cup B$
- $(A \cup B)' = A' \cap B'$
- None of above

Click to Save Answer & Move to Next Question

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 11 of 30 (Start time: 09:17:46 AM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d, e\}$ with topology $T = \{\phi, \{b\}, \{a, d\}, \{a, b, d\}, \{a, c, d, e\}, X\}$. The interior point of $A = \{a, b, c\}$ is:

Select the correct option

Reload Math Equations

- a
- d
- c
- b

Click to Save Answer & Move to Next Question

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 9 of 30 (Start time: 09:15:42 AM, 02 July 2021)

Total Marks: 1

More than one topological spaces can be defined on a single set.

Select the correct option

- True
- False

Click to Save Answer & Move to Next Question

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 13 of 30 (Start time: 09:19:51 AM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c\}$. *The following set is a topology on* X.

Select the correct option

Reload Math Equations

- {φ,{a},{b},X}
- {φ,{b},{c},X}
- {φ,{a},{b,c},X}
- none

Click to Save Answer & Move to Next Question

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 14 of 30 (Start time: 09:20:16 AM, 02 July 2021)

Total Marks: 1

Topology can be a useful tool in those problems where ..... study is more effective.

Select the correct option

- Qualitative
- Quantitative

Click to Save Answer & Move to Next Question

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 15 of 30 (Start time: 09:20:43 AM, 02 July 2021)

Total Marks: 1

The smallest topology one can define on some set is called:

Select the correct option

Reload Math Equations

- (●) indiscrete topology
- ( ) usual topology
- ( ) comparable topology
- ( ) discrete topology

Click to Save Answer & Move to Next Question

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 18 of 30 (Start time: 09:23:23 AM, 02 July 2021)

Total Marks: 1

Which of the following are NOT topologically equal?

Select the correct option

- A rectangle and a square
- A triangle and a rectangle
- A circle and a square
- A circle and a line

Click to Save Answer & Move to Next Question

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 17 of 30 (Start time: 09:21:56 AM, 02 July 2021)

Total Marks: 1

Which of the following is the basis for an upper limit topology on R (The Set of Real Numbers)?

Select the correct option

Reload Math Equations

- $B_{up} = \{[a, b) \mid a, b \in R\}$
- $B_{up} = \{[a, b] \mid a, b \in R\}$
- $B_{up} = \{(a, b] \mid a, b \in R\}$
- $B_{up} = \{(a, b) \mid a, b \in R\}$

Click to Save Answer & Move to Next Question

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 16 of 30 (Start time: 09:21:19 AM, 02 July 2021)

Total Marks: 1

Boundary point of a set is also called..... point.

Select the correct option

- limit
- frontier
- interior
- exterior

Click to Save Answer & Move to Next Question

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 19 of 30 (Start time: 09:24:15 AM, 02 July 2021)

Total Marks: 1

If τ is a topology on a non empty set X, then ..... of finite members of τ belongs to τ.

Select the correct option

Reload Math Equations

- intersection
- union

Click to Save Answer & Move to Next Question

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MTH634 Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 20 of 30 (Start time: 09:24:52 AM, 02 July 2021)

Total Marks: 1

An open disk in R^2 (Real plane) centered at $(x, y) \in R^2$, is the set of all points $(x', y') \in R^2$ such that

Select the correct option

Reload Math Equations

- $\{(x', y') | \sqrt{(x - x')^2 + (y - y')^2} = r,$ where r is a positive real number$\}$
- $\{(x', y') | \sqrt{(x - x')^2 + (y - y')^2} > r,$ where r is a positive real number$\}$
- $\{(x', y') | \sqrt{(x - x')^2 + (y - y')^2}\, a \leq r,$ where r is a positive real number$\}$
- $\{(x', y') | \sqrt{(x - x')^2 + (y - y')^2} < r,$ where r is a positive real number$\}$

Saving...

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Question # 21 of 30 (Start time: 09:25:35 AM, 02 July 2021) **Total Marks: 1**

Consider the following topology on $X = \{a, b, c, d, e\}$:

$$T = \{\emptyset, X, \{a\}, \{a, b\}, \{a, c, d\}, \{a, b, c, d\}, \{a, b, e\}\}$$

The member of the relative topology on $A = \{a, b\}$ is

Select the correct option

Reload Math Equations

- (●) {a, b}
- ( ) none
- ( ) {e}
- ( ) {c}

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MTH634:Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 24 of 30 (Start time: 09:28:10 AM, 02 July 2021)

Total Marks: 1

The interior of the four intervals $[a, b]$, (a, b), $(a, b]$, $[a, b)$ is

Select the correct option

Reload Math Equations

- [a,b]
- (a,b)

Saving

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MTH634:Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 23 of 30 (Start time: 09:27:16 AM, 02 July 2021) **Total Marks:** 1

Closure of a subset A of a topological space X is the intersection of all ..... of X containing A.

Select the correct option

- open subsets
- subsets
- closed subsets
- none of these

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MTH634:Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 22 of 30 (Start time: 09:26:32 AM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d\}$ and $\tau = \{\phi, \{a\}, \{a, b\}, X\}$ be a topology on X .then the open sets of X are :

Select the correct option

Reload Math Equations

- $\{a\},\ \{a, b\}$
- $\{b\},\ \{b, d\}$
- $\{d\}, \{a, d\}$
- $\{c\},\ \{c, d\}$

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MTH634:Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 25 of 30 (Start time: 09:29:03 AM, 02 July 2021)

Total Marks: 1

If in a topology τ on X, all subsets of X are called open and closed, then τ is called :

Select the correct option

Reload Math Equations

- (●) indiscrete space
- ( ) discrete space
- ( ) none
- ( ) metric space

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MTH634:Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 26 of 30 (Start time: 09:30:07 AM, 02 July 2021)

Total Marks: 1

A subset A of a topological space X is closed if and only if .....

Select the correct option

Reload Math Equations

- ● $A' \subset A$
- ○ $A \supset A'$
- ○ $A = A^c$
- ○ $A' = A^c$

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MTH634:Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 27 of 30 (Start time: 09:30:49 AM, 02 July 2021)

Total Marks: 1

In a topological space X let a set G is a neighborhood of each of its point. Then G is

Select the correct option

Reload Math Equations

- open
- closed

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MTH634:Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 30 of 30 (Start time: 09:33:23 AM, 02 July 2021)

Total Marks: 1

For any two topologies T1 and T2 defined on a non empty set then T1 is finer than T2 , iff for all U(open set) $\in$ T2 implies.....

Select the correct option

- $U \in T1$
- $U \notin T1$

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MTH634:Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 29 of 30 (Start time: 09:32:09 AM, 02 July 2021)

Total Marks: 1

Which of the following is the topology on real line induced by the set of all linear functions from R to R with lower limit topology is .....

Select the correct option

- Indiscrete
- discrete
- cofinite
- upper limit

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MTH634:Grand Quiz

Quiz Start Time: 09:02 AM, 02 July 2021

Question # 28 of 30 (Start time: 09:31:56 AM, 02 July 2021)

Total Marks: 1

Let X={1,2,3}, then P(X)=_____

Select the correct option

Reload Math Equations

- {φ,{1},{3},{1,2},{1,3},{1,2,3}}
- {φ,X}
- {φ,{1},{2},{3},{1,2},{1,3},{2,3}}
- {φ,{1},{2},{3},{1,2},{1,3},{2,3},X}

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 1 of 30 (Start time: 09:35:39 AM, 02 July 2021)

Total Marks: 1

If X is a finite set then co-finite topology on X is .....

Select the correct option

- indiscrete topology
- discrete topology
- lower limit topology
- none of these

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 2 of 30 (Start time: 09:36:13 AM, 02 July 2021)

Total Marks: 1

Let $X = R$ with usual topology and $A = (0, 4)$ then the limit point of A is:

Select the correct option

Reload Math Equations

- -2
- 2
- 5
- 6

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 3 of 30 (Start time: 09:37:38 AM, 02 July 2021)

Total Marks: 1

$\overline{(\bar{A})} =$

Select the correct option

Reload Math Equations

- ϕ
- $\bar{A}$
- None of above
- A

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 6 of 30 (Start time: 09:39:01 AM, 02 July 2021)

Total Marks: 1

A function $f : X \to Y$ from a topological space (X, T_X) to a topological space (Y, T_Y) is continuous iff for all $u \in T_Y$, there exist $v \in T_X$, such that

Select the correct option

Reload Math Equations

- $f^{-1}(u) = u$
- $f^{-1}(v) = v$
- $f^{-1}(v) = u$
- $f^{-1}(u) = v$

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 5 of 30 (Start time: 09:38:10 AM, 02 July 2021) **Total Marks:** 1

If τ is a topology on a non empty set X, then arbitrary ..... of members of τ belongs to τ.

Select the correct option

Reload Math Equations

- intersection
- union

Saving...

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 4 of 30 (Start time: 09:37:49 AM, 02 July 2021)

Total Marks: 1

If τ is a topology on a non empty set X, then ..... of finite members of τ belongs to τ.

Select the correct option

Reload Math Equations

- union
- intersection

Click to Save Answer & Move to Next Question

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 7 of 30 (Start time: 09:40:19 AM, 02 July 2021)

Total Marks: 1

Consider the discrete topology D on $X=\{a,b,c\}$. Find a subbase B for D which does not contain any singleton sets.

Select the correct option

Reload Math Equations

- $B=\{\{a,b\}, \{b,c\}\}$
- $B=\{\{a,b\}, \{b,c\}, \{c,a\}\}$

Click to Save Answer & Move to Next Question

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 8 of 30 (Start time: 09:42:14 AM, 02 July 2021) **Total Marks: 1**

Exterior point of a set is the ..... point of its complement

Select the correct option

- exterior
- boundary
- interior
- limit

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 9 of 30 (Start time: 09:42:41 AM, 02 July 2021)

Total Marks: 1

A function whose range is a singleton is called a ..... function.

Select the correct option

- continuous
- constant
- one to many
- Bijective

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 12 of 30 (Start time: 09:44:23 AM, 02 July 2021) **Total Marks:** 1

Consider the following topology on $X = \{a, b, c, d, e\}$:

$$T = \{\emptyset, X, \{a\}, \{a, b\}, \{a, c, d\}, \{a, b, c, d\}, \{a, b, e\}\}$$

The member of the relative topology on $A = \{a, b\}$ is

Select the correct option

Reload Math Equations

- ○ {e}
- ● {a, b}
- ○ none
- ○ {c}

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 11 of 30 (Start time: 09:43:50 AM, 02 July 2021) **Total Marks:** 1

For any two topologies T1 and T2 defined on a non empty set then T1 is finer than T2 , iff for all U(open set) ∈ T2 implies.....

Select the correct option

- $U \notin T1$
- $U \in T1$

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 10 of 30 (Start time: 09:43:14 AM, 02 July 2021)

Total Marks: 1

Let τ_1 and τ_2 be the two topologies on a non-empty set X. then τ_2 is finer than τ_1 iff.....

Select the correct option

Reload Math Equations

- $\tau_1 = \tau_2$
- $\tau_2 \subseteq \tau_1$
- $\tau_1 \subseteq \tau_2$
- None of these

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 13 of 30 (Start time: 09:44:42 AM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d, e\}$. *The following set* is a *topology* on X.

Select the correct option

Reload Math Equations

- {φ,{a},{b,c},{a,b,c},{d},X}
- {φ,{a,b},{a,c,d},X}
- {φ,{a},{a,b},{a,c,d},{a,b,c,d},{a,b,e},X}
- none

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 13 of 30 (Start time: 09:44:42 AM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d, e\}$. *The following set* is a *topology* on X.

Select the correct option

Reload Math Equations

- {φ,{a},{b,c},{a,b,c},{d},X}
- {φ,{a,b},{a,c,d},X}
- {φ,{a},{a,b},{a,c,d},{a,b,c,d},{a,b,e},X}
- none

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 14 of 30 (Start time: 09:45:28 AM, 02 July 2021)

Total Marks: 1

Let $X = \{2, 4, 6\}$.The following sets represents a topology on

X

.

Select the correct option

Reload Math Equations

- $\{\phi, X\}$
- All of these.
- $\{\phi, \{4\}, \{4, 6\}, X\}$
- $\{\phi, \{2\}, \{2, 4\}, X\}$

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 17 of 30 (Start time: 09:47:37 AM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d\}$. *The following set* is not *a topology on* X.

Select the correct option

Reload Math Equations

- {φ,{a},{a,b},X}
- {φ,{a},{a,c},X}
- {φ,{a},{a,d},X}
- {φ,{a},{b},X}

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Question # 16 of 30 (Start time: 09:46:49 AM, 02 July 2021)

Total Marks: 1

A subset U of R2 is open in R2 iff it contains an ..... at each of its points.

Select the correct option

- closed disk
- open disk
- open sphere
- closed interval

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 15 of 30 (Start time: 09:45:21 AM, 02 July 2021)

Total Marks: 1

The set of all open intervals of $\mathbb{R}$ is a topology on $\mathbb{R}$, called

Select the correct option

Reload Math Equations

- real topology
- discrete topology
- cofinite topology
- usual topology

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 18 of 30 (Start time: 09:47:52 AM, 02 July 2021)

Total Marks: 1

The collection τ of subsets of X consisting of the empty set ϕ and all subsets of X whose complements are finite is called :

Select the correct option

Reload Math Equations

- discrete topology
- indiscrete topology
- cofinite topology
- none

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 19 of 30 (Start time: 09:48:25 AM, 02 July 2021)

Total Marks: 1

Let $X = \mathbb{R}$ with usual topology and $A = (0, 3)$. *The limit point of* A is:

Select the correct option

Reload Math Equations

- -0.5
- 5
- 1.5
- 4

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 20 of 30 (Start time: 09:48:42 AM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d\}$ *and* $\tau = \{\phi, \{c\}, \{a, c\}, \{b, c, d\}, X\}$ *be a topology on* X. The closed set in X is:

Select the correct option

Reload Math Equations

- {d}
- {c}
- {b,d}
- none

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 23 of 30 (Start time: 09:50:56 AM, 02 July 2021)

Total Marks: 1

The neighborhood system of a point p in an indiscrete space X is

Select the correct option

Reload Math Equations

- ○ $\emptyset$
- ○ none
- ○ X
- ● p

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 22 of 30 (Start time: 09:50:30 AM, 02 July 2021) **Total Marks:** 1

Closure of a subset A of a topological space X is the intersection of all ..... of X containing A.

Select the correct option

- none of these
- open subsets
- closed subsets
- subsets

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 21 of 30 (Start time: 09:49:34 AM, 02 July 2021)

Total Marks: 1

Let (X, τ) be the topological space and $A, B \subset X$.If $A \subset B$ then:

Select the correct option

Reload Math Equations

- $\bar{A} \subset B$
- $A \subset B$
- $\bar{A} \subset \bar{B}$
- None of above

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 24 of 30 (Start time: 09:52:31 AM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d\}$ and $\tau = \{\phi, \{a\}, \{a, b\}, X\}$ be a topology on X.then the open sets of X are :

Select the correct option

Reload Math Equations

- $\{d\}, \{a, d\}$
- $\{b\}, \{b, d\}$
- $\{c\}, \{c, d\}$
- $\{a\}, \{a, b\}$

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 25 of 30 (Start time: 09:52:43 AM, 02 July 2021) **Total Marks:** 1

If one shape can be deformed in another shape then topologically they are considered to be _____

Select the correct option

- unique
- the same
- dissimilar
- partially equal

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 24 of 30 (Start time: 09:52:31 AM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d\}$ and $\tau = \{\phi, \{a\}, \{a, b\}, X\}$ be a topology on X.then the open sets of X are :

Select the correct option

Reload Math Equations

- $\{d\}, \{a, d\}$
- $\{b\}, \{b, d\}$
- $\{c\}, \{c, d\}$
- $\{a\}, \{a, b\}$

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 27 of 30 (Start time: 09:53:19 AM, 02 July 2021)

Total Marks: 1

Let $X = R$ be a usual topology and consider $C = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4} \ldots\ldots\ldots\ldots\ldots\}$ .then which of the following is not limit point of :

Select the correct option

Reload Math Equations

- 0
- 1/2
- 1
- Both (b) and (c)

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 26 of 30 (Start time: 09:53:03 AM, 02 July 2021)

Total Marks: 1

In a topological space, the interior of a set A is the union of all

Select the correct option

Reload Math Equations

- open subsets of A
- closed subsets of A

Saving...

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 25 of 30 (Start time: 09:52:43 AM, 02 July 2021)

Total Marks: 1

If one shape can be deformed in another shape then topologically they are considered to be _____

Select the correct option

- unique
- the same
- dissimilar
- partially equal

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 28 of 30 (Start time: 09:54:25 AM, 02 July 2021)

Total Marks: 1

..... is the largest topology containing both T1 and T2.

Select the correct option

- Indiscrete topology
- Cofinite topology
- T1 union T2
- T1 intersection T2

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 29 of 30 (Start time: 09:55:02 AM, 02 July 2021)

Total Marks: 1

Let X={1,2,3}, then P(X)=_____

Select the correct option

Reload Math Equations

- {φ,{1},{2},{3},{1,2},{1,3},{2,3},X}
- {φ,X}
- {φ,{1},{3},{1,2},{1,3},{1,2,3}}
- {φ,{1},{2},{3},{1,2},{1,3},{2,3}}

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MTH634:Grand Quiz

Quiz Start Time: 09:35 AM, 02 July 2021

Question # 30 of 30 (Start time: 09:55:36 AM, 02 July 2021)

Total Marks: 1

If in a topology τ on X, all subsets of X are called open and closed, then τ is called :

Select the correct option

Reload Math Equations

- (●) indiscrete space
- ( ) none
- ( ) metric space
- ( ) discrete space

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 3 of 30 (Start time: 09:57:11 AM, 02 July 2021)

Total Marks: 1

For any two topologies T1 and T2 defined on a non empty set then T1 is finer than T2 , iff for all U(open set) ∈ T2 implies.....

Select the correct option

- ○ U ∉ T1
- ● U ∈ T1

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 2 of 30 (Start time: 09:56:57 AM, 02 July 2021)

Total Marks: 1

Let $X = \mathbb{R}$ with usual topology and $B = [1, 2)$. Then interior of a set B is

Select the correct option

Reload Math Equations

- none
- 1
- 3/2
- 2

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 1 of 30 (Start time: 09:56:33 AM, 02 July 2021)

Total Marks: 1

Which of the following is true for a subset A of a Topological space X.

Select the correct option

Reload Math Equations

- $X = A^0 \cup Ext(A) \cup bd(A)$
- $X = A^0 \cap Ext(A) \cup bd(A)$
- $X = A^0 \cap Ext(A) \cap bd(A)$
- $X = A^0 \cup Ext(A) \cap bd(A)$

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 4 of 30 (Start time: 09:57:16 AM, 02 July 2021)

Total Marks: 1

If τ is a topology on a non empty set X, then ..... of finite members of τ belongs to τ.

Select the correct option

Reload Math Equations

- intersection
- union

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Question # 5 of 30 (Start time: 09:57:22 AM, 02 July 2021)

Total Marks: 1

Let $X = \{2, 4, 6\}$.The following sets represents a topology on

$$X$$

.

Select the correct option

Reload Math Equations

- All of these.
- $\{\phi, \{2\}, \{2, 4\}, X\}$
- $\{\phi, \{4\}, \{4, 6\}, X\}$
- $\{\phi, X\}$

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 6 of 30 (Start time: 09:57:28 AM, 02 July 2021)

Total Marks: 1

A function $f : X \to Y$ from a topological space (X, T_X) to a topological space (Y, T_Y) is continuous iff for all $u \in T_Y$, there exist $v \in T_X$, such that

Select the correct option

Reload Math Equations

- $f^{-1}(u) = v$
- $f^{-1}(v) = v$
- $f^{-1}(u) = u$
- $f^{-1}(v) = u$

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 9 of 30 (Start time: 09:58:33 AM, 02 July 2021)

Total Marks: 1

Closure of A is the smallest closed superset of A.

Select the correct option

Reload Math Equations

- False
- True

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 10 of 30 (Start time: 09:59:04 AM, 02 July 2021)

Total Marks: 1

Let $X = \{2, 4, 5, 7\}$. *The following set represents a topology on* X.

Select the correct option

Reload Math Equations

- {φ,{2},{4,5},X}
- {φ,{2},{4,5,7},X}
- {φ,{2},{4},{5},X}
- {φ,{2},{5,7},X}

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 11 of 30 (Start time: 09:59:26 AM, 02 July 2021)

Total Marks: 1

The smallest topology one can define on some set is called:

Select the correct option

Reload Math Equations

- discrete topology
- indiscrete topology
- usual topology
- comparable topology

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 12 of 30 (Start time: 09:59:33 AM, 02 July 2021)

Total Marks: 1

Exterior point of a set is the ..... point of its complement.

Select the correct option

- boundary
- exterior
- interior
- limit

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 15 of 30 (Start time: 10:01:07 AM, 02 July 2021)

Total Marks: 1

The boundary of the four intervals $[a, b]$, (a, b), $(a, b]$, $[a, b)$ is

Reload Math Equations

Select the correct option

- {a, b}
- (a, b)

Saving...

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 14 of 30 (Start time: 10:00:16 AM, 02 July 2021) **Total Marks:** 1

The set of all singletons forms a basis for ..... topology.

Select the correct option

- indiscrete
- lower limit
- co finite
- discrete

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 8 of 30 (Start time: 09:58:17 AM, 02 July 2021)

Total Marks: 1

_____ is the weakest topology on set X .

Select the correct option

Reload Math Equations

- Cofinite Topolgy
- Indiscrete Topology
- Discrete Topology
- Noe of above

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 7 of 30 (Start time: 09:58:01 AM, 02 July 2021) **Total Marks:** 1

The set of all open disks forms a usual basis for the usual topology defined on ___

Select the correct option

- R3(space)
- Rn(n tuple)
- R(Real line)
- R2(Real plane)

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 13 of 30 (Start time: 09:59:41 AM, 02 July 2021) **Total Marks: 1**

A map 'f' from a topological space X to a topological space Y is _____ if and only if the image of each element of a basis for TX(Topology on X) is open subset of Y.

Select the correct option

- surjective
- bijective
- injective
- an open map

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 16 of 30 (Start time: 10:01:14 AM, 02 July 2021)

Total Marks: 1

If B is a basis for a topology T on a set X then ..... element of T can be written as union of the elements of B.

Select the correct option

- one
- every
- some
- no

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 17 of 30 (Start time: 10:02:17 AM, 02 July 2021)

Total Marks: 1

Let $X = \{a, b, c, d\}$. *The following set* is *a topology on* X.

Select the correct option

Reload Math Equations

- {φ,{b},{c},X}
- {φ,{b},{a,b},{c},X}
- none
- {φ,{b,d},{a,b,c},{b},X}

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 18 of 30 (Start time: 10:03:08 AM, 02 July 2021)

Total Marks: 1

Let X be a topological space and A, a subset of X, then

Select the correct option

Reload Math Equations

- $A' = A \cup \bar{A}$
- $\bar{A} = A \cup A'$
- $\bar{A} = A \cap A'$
- none

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 21 of 30 (Start time: 10:04:56 AM, 02 July 2021) **Total Marks:** 1

Let $X = \{1, 2, 3\}$ and $\tau = \{\phi, \{1\}, \{2\}, \{1, 2\}, X\}$ then (X, τ) as a pair will be called .....

Select the correct option

Reload Math Equations

- Indiscrete topology
- None of these
- Topological space
- Discrete topology

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 20 of 30 (Start time: 10:04:18 AM, 02 July 2021)

Total Marks: 1

If ζ is a subbase for topologies T and T^*, then

Select the correct option

Reload Math Equations

- $T \neq T^*$
- $T = T^*$

Click to Save Answer & Move to Next Question

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 19 of 30 (Start time: 10:04:07 AM, 02 July 2021)

Total Marks: 1

Derived set of empty set .....

Select the correct option

- All of above
- 1
- Empty
- Non empty

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 22 of 30 (Start time: 10:05:19 AM, 02 July 2021)

Total Marks: 1

In a topological space X and a subset A of X , A is dense in X if and only if.....

Select the correct option

Reload Math Equations

- $\bar{A} = X$
- $\bar{X} = X$
- $\bar{A} \subset X$
- $\bar{A} = A$

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 25 of 30 (Start time: 10:06:23 AM, 02 July 2021)

Total Marks: 1

In a discrete space X, the derived set A' of any subset A of X is _____ .

Select the correct option

- empty set
- X
- A complement
- A itself

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 27 of 30 (Start time: 10:06:59 AM, 02 July 2021)

Total Marks: 1

Consider the discrete topology D on $X=\{a,b,c\}$. Find a subbase B for D which does not contain any singleton sets.

Select the correct option

Reload Math Equations

○ $B=\{\{a,b\},\{b,c\},\{c,a\}\}$

● $B=\{\{a,b\},\{b,c\}\}$

Click to Save Answer & Move to Next Question

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 30 of 30 (Start time: 10:08:27 AM, 02 July 2021) **Total Marks:** 1

Which of the following is the smallest closed super set of a subset A of a topological space X?

Select the correct option

- Derived set of A
- Complement of A
- open sets of A
- Closure of a set A

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 29 of 30 (Start time: 10:07:12 AM, 02 July 2021)

Total Marks: 1

Let A be the collection of all closed subset of a topological space (X, τ) then A is closed under _____set union .

Select the correct option

Reload Math Equations

- Finite
- Infinite

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MTH634:Grand Quiz

Quiz Start Time: 09:56 AM, 02 July 2021

Question # 28 of 30 (Start time: 10:07:04 AM, 02 July 2021)

Total Marks: 1

Let (X, τ) be the topological space and $A, B \subset X$.If $A \subset B$ then:

Select the correct option

Reload Math Equations

- None of above
- $\bar{A} \subset B$
- $A \subset B$
- $\bar{A} \subset \bar{B}$

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